

Simplified Trusted Nodes using Twin Field QKD

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Abstract—QKD allows for key establishment between two parties, without the need for computational assumptions. However, over long distances it becomes inefficient, or impractical, unless a quantum network is used. Normally, trusted node networks are used today, however these require each trusted node to be capable of performing high-speed error correction and privacy amplification. Simplified trusted nodes (STNs) do not have this requirement, however they introduce additional challenges in terms of “cascading noise” in the link, limiting the total number of hops in a network path. We attempt to mitigate this problem by combining the STN architecture with twin-field QKD (TF-QKD). Our solution maintains the computational benefits of STN systems, while also helping to alleviate their distance limitations.

Index Terms—Quantum Key Distribution, Quantum Networking, Security

I. INTRODUCTION

Quantum key distribution (QKD) holds great potential, yet several shortcomings must be addressed before it can see wide-scale adoption. Perhaps the most important issue to address are distance limitations. Due to the exponential loss over fiber, the need to send single photons, and the inability to amplify a quantum signal, QKD in practice is generally limited in distance. Quantum networks, and new quantum protocols, can both help to overcome this limitation.

While quantum networks consisting of true quantum repeaters are still being developed, quantum networks consisting of *trusted node* (TN) technology is very much a real technology, today. Several metropolitan areas are building, or have built, trusted node networks to support QKD (see [1]–[5] for just a few examples); trusted node networks have also been used to link data centers [6]. Trusted nodes work by establishing a shared secret key with its adjacent network nodes (which are either other trusted nodes, or end-users such as Alice and Bob). Consider the simplest network structure, a single path consisting of Alice and Bob at the end-points, connected to one another through a path of p trusted nodes, denoted $\mathcal{T}^{(1)}$ through $\mathcal{T}^{(p)}$. Alice will establish a secret key, using some link-level QKD protocol (e.g., BB84), with $\mathcal{T}^{(1)}$. Meanwhile, $\mathcal{T}^{(1)}$ will also establish, an independent secret key with $\mathcal{T}^{(2)}$ using QKD. And so on; thus $p + 1$ secret keys are established, one for each link of the path. $\mathcal{T}^{(i)}$ will

send the XOR sum of its two secret keys to Bob (effectively encrypting, through OTP encryption, one secret key using the other). This will allow Bob to faithfully reconstruct Alice’s secret key established originally with $\mathcal{T}^{(1)}$.

While the above process works, each TN is required to run a full QKD stack *twice*: once with its left neighbor and once with its right. This means each TN must run a quantum communication stage to establish raw keys, then process them through error correction and privacy amplification to establish secret keys. Error correction can be a computationally expensive process, due to the large block sizes of QKD, and each TN must run this process twice for every one secret key that Alice and Bob wish to establish. In a large scale network, which must serve multiple pairs of end users, this can be a bottleneck.

To mitigate this issue, the *simplified trusted node* (STN) architecture was introduced in [7]. Here, an STN performs the same quantum communication process as a TN, establishing raw keys with its neighbors, but it never performs error correction and privacy amplification. Nor does it perform any error sampling. Instead, it simply sends the XOR sum of its two raw keys to Bob, allowing him to reconstruct a noisy version of Alice’s raw key established with $\mathcal{T}^{(1)}$. Alice and Bob will then run the final error correction and privacy amplification stages, which they would normally have to do anyway. However, the STN network is immediately free to service other users. Due to the cascading nature of link-level noise in an STN network, overall key-sizes may be smaller in a single session, however, STN networks may be cheaper to deploy or can be used in scenarios where limited computation is available. It also may be deployed if TNs “break down” partially, and can no longer perform the expensive classical post processing. Thus, studying STN architectures is an important research problem.

As discussed, errors cascade in the network chain, leaving Alice and Bob with a noisy raw key, the noise of which is larger than the error in a single link. This limits the total number of STNs in a single chain (there is no theoretical limit to the number of TNs in a chain, since each one performs error correction and privacy amplification on each link). Ultimately, this could limit the total distance supported by an STN network (though still outperforming a single link). Another challenge when working with STNs as opposed to TNs, is that whenever one wishes to use a different link-level QKD protocol, one requires a new proof of security. This is due to the fact that Eve, the adversary, may influence each link simultaneously,

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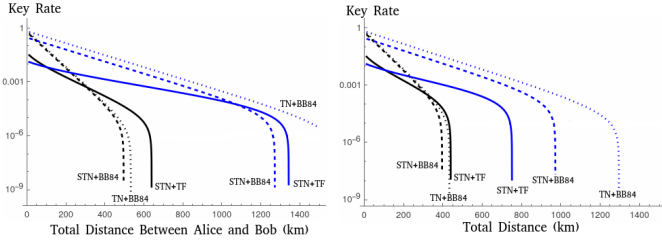


Fig. 1. Asymptotic performance of our proposed STN+TF (Solid lines), compared with prior work using the BB84 protocol for both STN networks (STN+BB84, Dashed lines) from [7] and standard trusted node networks (TN+BB84, Dotted lines). Black: $p = 1$ (one STN or TN); Blue: $p = 5$ (five STNs or TNs in the chain). For STN+BB84 and TN+BB84, we assume a low link level depolarizing noise of 1%. **Left:** A detector dark count rate of 10^{-6} and an efficiency of 80%. **Right:** Same efficiency, but a higher dark count rate of 10^{-5} . We see that when the dark count rate is 10^{-6} , STN+TF outperforms STN+BB84, but when it is higher (Right), STN+BB84 outperforms at these settings.

creating arbitrary correlations, and entangling herself with the entire chain. A TN network runs error correction and privacy amplification on each link and so does not share this concern.

Since STNs hold several potential advantages over TNs, it's important to try and find ways to improve their performance over larger distances. Our work proposes the use of Twin Field QKD (TF-QKD) as the underlying link-level QKD protocol. TF-QKD was introduced in [8]. The specific protocol we consider is from [9]. TF-QKD's key-rate scales with $O(\sqrt{\eta})$, where η is the link transmissivity and can even beat the repeater-less PLOB bound [10]. We combine TF-QKD with the STN architecture (denoting the overall protocol STN+TF), derive an information theoretic proof of security for the new protocol, and show that it can outperform standard STN networks using BB84 (which we denote STN+BB84), allowing for greatly improved distances. All prior STN work, to our knowledge, only considered BB84 as the underlying link level protocol. Our work is the first to consider an alternative link level protocol, and is the first to address the distance limitations of STN networks.

This paper is an extended abstract of a full version of the paper currently under development. Here, we discuss briefly the protocol and asymptotic security results. In the future full version of this paper, we anticipate deriving a full finite key security proof, and a more rigorous evaluation and comparison to TN network architectures and STN+BB84.

Results: Using techniques from [7], [11], [12], we develop an asymptotic key-rate expression for STN+TF. Our expression takes the form $r_\infty = p_a(1 - h(Q_X) - h(Q_Z))$, where Q_X is the observed *total* X basis noise in the STN chain, while Q_Z is the observed Z basis noise between Alice and Bob (i.e., the total Z basis noise in the chain, after STN runs). Note that since the TF-QKD protocol we are using has an asymmetry in the two bases [9], we need to separate the parameters. The value p_a is the probability that any particular round is accepted (not discarded) by the underlying TF-QKD protocol.

To evaluate, we simulate the expected link level noise of

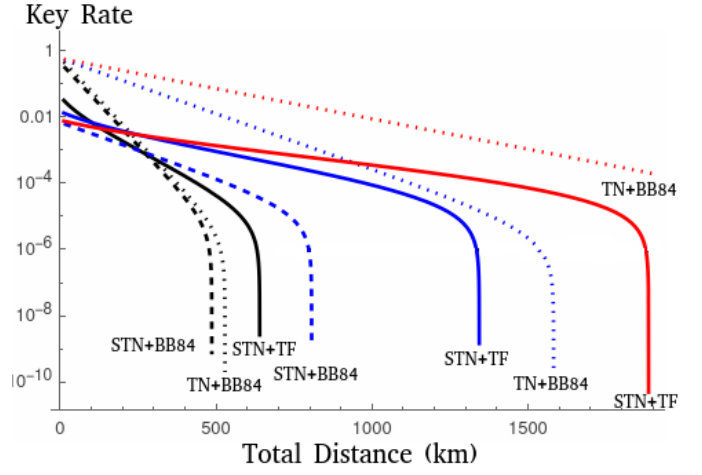


Fig. 2. Comparing STN+TF (Solid) with STN+BB84 (Dashed) and TN+BB84 (Dotted). Black: one STN/TN; Blue: five STNs/TNs; Red: ten STNs/TNs. Here the dark count rate is 10^{-6} while the detector efficiency is 80%. We set the link-level depolarizing noise to 2% for STN+BB84 and TN+BB84. We see that, here, STN+TF can substantially outperform STN+BB84 as p increases. In fact, when $p = 10$ (Red), STN+BB84 cannot even establish a secret key, so it is not plotted.

the underlying TF-QKD protocol, and the expected value of p_a , using results from [9]. We then estimate the expected total noise Q_X (similarly for Q_Z) using the expression:

$$\mathcal{E}(Q_X; p) = \sum_{i=0}^{\lfloor p/2 \rfloor} \binom{p+1}{2i+1} Q_X^{2i+1} (1 - Q_X)^{p-2i} \quad (1)$$

which would be the expected total noise if the links were independent. Of course, in practice, the links may not be independent, and our security proof handles that case, one just requires the observed noise levels in the chain. We are using Equation 1 only for evaluation purposes. Our results are shown in Figures 1 and 2 which reports the maximal distances between Alice and Bob, for various number of STNs p , before the key-rate drops to zero. We assume the total distance between Alice and Bob is D km, and thus the total link distance for one hop of the network is $d = D/(p+1)$. We assume the transmissivity of a single link, then, is $\eta = 10^{-.2d/10}$. Note that our work outperforms the prior work STN+BB84, except when the dark-count rate of the twin field units are high. When $p = 1$, we can even outperform standard trusted node networks using BB84. Of course, we would never outperform standard TN networks using TF-QKD *in terms of secret key bits per network rounds*; however as discussed, the advantage to STNs are their potentially cheaper cost and overall efficiency. They may also be faster when serving multiple end users, as an STN network does not need to perform any classical computation after the quantum portion of the protocol is finished, leading to overall faster key generation rates [13], [14] which we are not considering here. Evaluating this to perform an accurate comparison of the relative costs of each network architecture, is an important future research problem.

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